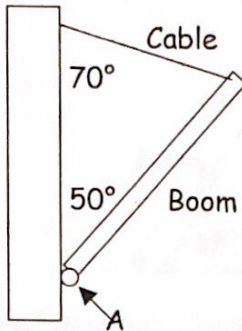


Equilibrium Worksheet#3

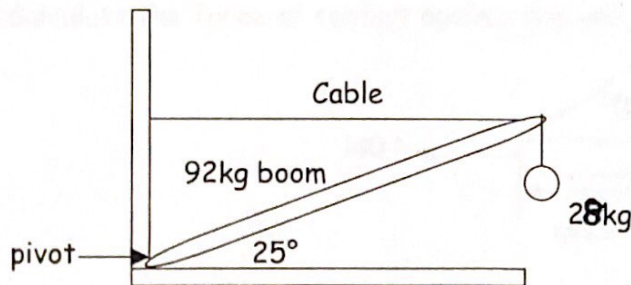
(name) _____

1. The following diagram shows a uniform boom of length 2.6 m and mass 53 kg. The boom is hinged to the wall at point A and supported by a cable as shown. What is the vertical component of the force acting at point A?



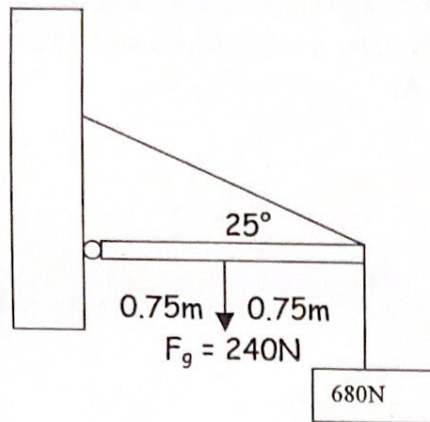
F = _____

2. A uniform boom of length 3.6 m and mass 92 kg supports a load of 28 kg as shown. What is the tension in the horizontal cable supporting the boom?



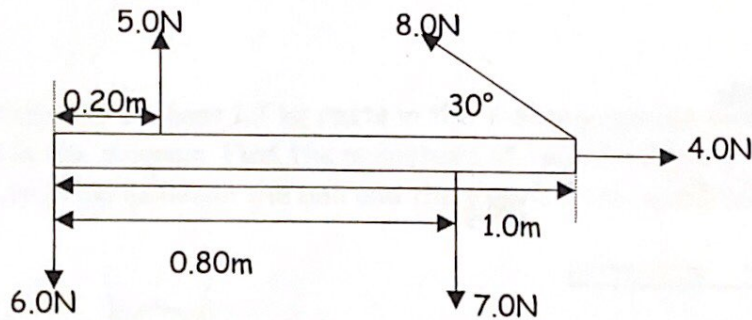
F = _____

3. What is the tension in the cable shown in the following diagram?



F = _____.

4. A bar 1.0 m long has five forces acting on it as shown in the following diagram. What are the magnitude, direction, and location of the single force required to produce static equilibrium? The weight and thickness of the bar are not significant.



F = _____.

Answers: 1. 440N 2. $1.6 \times 10^3 \text{ N}$
~~4. 496N, 54° N of E, 0.15m from left end~~ 3. $1.9 \times 10^3 \text{ N}$
 5.0N